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Large language model agents in a collective-risk social dilemma: capability enables risk sensitivity but does not confer it

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Collective-risk social dilemmas, of which climate change is the archetype, reward a group only if its pooled contribution crosses a threshold before a probabilistic catastrophe, and in the canonical human experiment cooperation rises steeply with the stated risk. As large language models (LLMs) are deployed as autonomous agents and used as surrogates for human decision-makers, we ask whether they reproduce that risk sensitivity. Thirteen instruction-tuned models played the dilemma as six-agent groups: seven open-weight models from 7 to 72 billion parameters, and six commercial models covering four providers at the top capability tier and two of them at a cheap tier as well, crossing catastrophe risk ($p = 0.1, 0.5, 0.9$) with language and, in a second study, with the prosocial composition of the group. Eleven of the thirteen ignore the risk, and for them the null is not a ceiling artefact: where target-reach is genuinely uncertain, a ninefold rise in the probability of ruin moves contributions by $+0.97$ of a possible 240 (95% CI $[-2.2, +4.1]$, $P = 0.54$), whereas three manipulations delivered through the prompt rather than through the incentive structure each move the same quantity by one to two orders of magnitude more: printing the running pool total (up to 95.3 points), changing group composition (31.8–112.4 points), and switching language (146.8 points, inverting a 100% reach rate to 0%). Two models overturn any universal reading of that null. Gemini-3.1-Pro and GPT-5.6-sol withhold everything at $p = 0.1$ and contribute exactly the target above it, a risk effect of $+120.0$ and $+118.2$ points that lifts realised payoff from 20.0 to 32.0 per agent. Which models do this is predicted by neither provider nor tier alone: four top-tier models from four laboratories divide two against two, no cheap model ever solves the game, and two members of one family and generation differ by the full width of the board. Toggling explicit reasoning within one base model buys 105 points of thrift and no risk sensitivity at all. Even the responsive models are not human-like, stepping from 0% to 100% reach at the expected-value threshold against a human curve that climbs gradually. Behavioural claims about LLM agents need prompt-side controls, and a panel of models, before they are attributed to preferences.

The prevention of dangerous climate change is a collective-risk social dilemma: many actors must jointly invest enough to keep a shared catastrophe from occurring, yet each actor is individually tempted to free-ride on the investment of others [1–3]. In laboratory social dilemmas, humans sustain cooperation through reciprocity and costly punishment and respond to the material stakes at hand [4]. Laboratory versions of this dilemma have a sharp empirical signature. When Milinski and colleagues had groups of six people play a ten-round threshold game in which failing to reach the target triggered the probabilistic loss of everything they had kept, the fraction of groups that reached the target rose steeply with the announced risk of loss, from essentially zero when the risk was low to about half when the risk was high [1]. Humans, in other words, cooperate more when more is at stake.

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1. Introduction

Large language models (LLMs) are now used both as autonomous agents that negotiate, plan and transact on a user's behalf, and as inexpensive surrogates for human participants in the social and economic sciences [5–7]. A growing literature places these models inside classic games and finds that they can cooperate, reciprocate and punish, but often in ways that diverge from human play, and not always in the same direction: some models are strikingly prosocial and forgiving in repeated dilemmas [8, 9], whereas others retaliate unforgivingly after a single defection [10] or fail to sustain a shared resource without institutional scaffolding [11]. If such agents are to be trusted in, or used to forecast, high-stakes collective decisions, we need to know how they behave when the dilemma is structured by *risk* rather than by a fixed payoff matrix, because it is precisely the response to risk that gives the human collective-risk dilemma its policy relevance.

This question is entangled with a methodological one that recurs throughout the study of LLM behaviour. When a model deviates from the human pattern, the deviation admits two very different readings. It may be a genuine behavioural disposition, in which the model understands the situation and nonetheless chooses differently, or it may be an artefact of comprehension, in which the model has simply misread the rules, the history or the current state. These readings have opposite implications for whether scaling, prompting or fine-tuning will change the behaviour, yet most behavioural studies of LLMs do not measure comprehension directly and therefore cannot distinguish them. Recent work on the prisoner's dilemma has begun to close this gap by interleaving behavioural play with a battery of understanding checks, asking the model to report the rules, the game history and the current state, and scoring the answers against ground truth [8].

A third problem is specific to reporting null results in threshold games, and it is the one that most limits what can be concluded from a study of this kind. When agents cooperate enough to clear the threshold in essentially every game, the outcome variable is censored at its ceiling: a risk response of the human size could not appear in it even if it existed. A flat reach rate is therefore weak evidence on its own. The remedy is to find conditions in which the outcome is genuinely uncertain and to test the risk response there, and a manipulation that varies how cooperative the group is provides exactly such conditions.

A fourth problem is one of coverage, and our own results are what forced it on us. Evaluations of LLM behaviour are expensive, so they are usually run on one model, or on one family, and the conclusion is then stated of LLMs in general. That inference is only safe if the behaviour in question is a property of the technology rather than of the individual system. Whether risk sensitivity is such a property is an empirical question, and it can only be answered by holding the game fixed and varying the model along the two axes that matter commercially: which laboratory built it, and how much capability was bought.

Here we bring these questions together for the collective-risk social dilemma. We instantiated the Milinski experiment with LLM agents, replacing each of the six human players with an instruction-tuned model, and swept the announced catastrophe risk across the same three levels used with humans. To ask whether behaviour is bounded by capability, we placed seven open-weight models spanning three families and an order of magnitude in size, from 7 to 72 billion parameters, in the identical protocol, and then extended the panel to six commercial models covering the top capability tier at four providers and, at two of them, a cheap tier as well, so that the effect of capability could be read within a family as well as across the panel. To separate disposition from comprehension, we adapted the understanding-check methodology of Fontana et al. [8] from the two-player prisoner's dilemma to the N-player threshold game, probing each model *in situ* on the very prompts from which it was making decisions, with all answers graded against the game engine's ground truth. To break the ceiling and obtain uncensored conditions, we ran a second study that varied the prosocial composition of the group, assigning between zero and six of the six agents a selfish persona. Finally, because these models are deployed multilingually and the collective-risk framing has an intrinsic numerical structure, we ran every condition in both English and Vietnamese.

Four findings organise the paper. First, eleven of the thirteen models show no consequential response to risk, and for them the null is not an artefact of censoring: in the conditions where reach is genuinely uncertain, a ninefold increase in catastrophe risk moves contributions by an amount statistically indistinguishable from zero, while three manipulations that carry no incentive information whatsoever, printing the running pool total, changing who is in the group, and changing the language, each move contributions by one to two orders of magnitude more. Second, two models break that null completely, playing the expected-value solution exactly: they contribute nothing at all when the catastrophe is unlikely and exactly the target when it is not, and are paid 60% more for it. Third, which models do this is not predicted by capability tier or by provider. Four top-tier models from four providers divide two against two; no cheap model ever solves the game; and the sharpest contrast in the study is between two members of the same family and generation that differ only in tier. Toggling explicit reasoning within one base model changes how economically it reaches the target without changing whether it asks if the target is worth reaching. Fourth, even the two responsive models are not human: their reach rate is a step function at the expected-value threshold, more decisive than the graded human curve rather than blinder than it. Throughout we are careful about a threat that applies to this literature generally and to our design in particular: our prompt names the equal-split solution, and some models reproduce it exactly, so cooperative-looking behaviour cannot be cleanly separated from compliance with a supplied focal point.

2. Material and methods

(a) The collective-risk social dilemma

We reproduced the parameters of the canonical human experiment [1]. Six players each began with an endowment of €40. Over ten rounds, each player privately and simultaneously chose to contribute 0, 2 or 4 to a common pool. If the pooled total reached the target of €120 by the end of the game, players kept whatever they had not contributed; if the target was missed, then with probability p, the catastrophe risk, every player lost all remaining private money. The maximum achievable pool is $6 \times 4 \times 10 = 240$, so the target of 120 can be met if players contribute, on average, half of the maximum. We used the three risk levels of the original study, $p \in \{0.1, 0.5, 0.9\}$. For humans, the fraction of groups reaching the target rose with risk, from 0% at $p = 0.1$ to 10% at $p = 0.5$ to 50% at $p = 0.9$ [1]; this monotone increase is the behavioural benchmark against which we compare the models.

The same parameters also define a benchmark of a second kind, which becomes essential once some models start to exploit it. Contributing the target share costs an agent 20 with certainty. Contributing nothing keeps 40 but exposes it to the lottery, for an expected $(1 - p) \times 40$: that is 36 at $p = 0.1$, 20 at $p = 0.5$ and 4 at $p = 0.9$. Under risk neutrality the dilemma therefore has a sharp structure that humans do not follow: withholding everything is strictly better at $p = 0.1$, the two are exactly indifferent at $p = 0.5$, and cooperating is strictly better at $p = 0.9$. We refer to this as the expected-value solution and use it as a normative reference point, not as a claim about what any agent computes.

(b) Language-model agents and the game loop

Each human player was replaced by an instruction-tuned LLM. We built a configuration-driven engine, in the style of agent-based game frameworks for LLMs [12], in which the six agents in a group were homogeneous (the same model) and, in the baseline study, anonymous (neutral identifiers, no assigned personas). On each round, every agent received a natural-language prompt containing the rules of the game and the complete round-by-round history, and was asked to end its reply with a line of the form `CONTRIBUTION:` c with $c \in \{0, 2, 4\}$, which we parsed deterministically. Agents therefore had full-history memory and a neutral framing that neither encouraged nor discouraged cooperation. One property of this prompt matters for every interpretation that follows and we state it plainly: in describing the target it also names the equal-split solution, telling the agent that €120 corresponds to “an average of 2 per player per round” and adding the worked example that contributing 2 every round leaves 20 at the end. The prompt therefore supplies a risk-independent focal action. It is faithful to the information a human subject could trivially compute, but it means that an agent contributing 2 may be solving the dilemma or may simply be following the number in front of it, and our design cannot separate the two. We return to this among the limitations in §4. The structure of the decision prompt is summarised in figure 1(a).

Catastrophes were resolved by a single end-of-game lottery. That lottery is the weakest part of the implementation. It draws one uniform variate per repetition, seeded from the repetition index alone, so the same draws are shared across every model, language and risk level: the study contains ten independent lottery outcomes in total, not one per game. All ten happen to fall below 0.5 (mean 0.276), so at $p = 0.5$ and $p = 0.9$ every game that missed the target was recorded as a catastrophe, while at $p = 0.1$ two of the ten draws triggered one. Realised catastrophe rates are therefore an artefact of the draw schedule and we exclude them from all inference; they are reported only descriptively, and where we quote a realised payoff we say what the draw schedule contributed to it.

(c) The model panel

The panel was assembled in two stages, and the second stage exists because of what the first one showed.

The open-weight stage comprised seven models spanning three families and roughly an order of magnitude in size: Qwen2.5 at 7B, 32B and 72B [13]; Gemma-2 at 9B and 27B [14]; and Llama-3.1 at 8B and 70B [15]. These were served locally with vLLM at a 4096-token context and decoded at temperature 0.7 with a 512-token cap; the two 70–72B models were served with activation-aware weight quantisation. Sampling was seeded per turn from the experiment seed, the repetition, the round and the agent index, so that runs are reproducible and independent across repetitions while using common random numbers across the risk and language conditions, isolating the manipulated factor.

Because a pattern shared by seven models of broadly common lineage might reflect that lineage rather than instruction-tuned LLMs generally, we extended the panel to commercial models reached through hosted chat-completions interfaces. We chose them so that capability can be read within a provider and not only across the panel: at the top tier, Gemini-3.1-Pro (Google), GPT-5.6-sol (OpenAI), Claude-Opus-5 (Anthropic) and Grok-4.20 (xAI); at the cheap tier, Gemini-3.1-Flash-Lite (Google) and GPT-5.4-nano (OpenAI). The grid is therefore complete in providers at the top tier and paired in tier at two of the four providers; budget, not design, is the reason Anthropic and xAI are represented at one tier only. Grok-4.20 was run in both its reasoning and its non-reasoning configuration, giving a within-model contrast that isolates explicit deliberation, so the six commercial models yield seven configurations and the full panel comprises thirteen models in fourteen configurations. Every commercial model played the same full factorial of three risk levels by two languages with ten independent repetitions, giving 60 games and $3{,}600$ agent-turns per configuration, except GPT-5.4-nano, which was run earlier at five repetitions (30 games). Contributions were parsed without failure everywhere in the study: the parse-failure rate is 0% for all fourteen configurations, so every game contributed valid data.

This commercial arm is a generalisation test rather than a like-for-like comparison, and three differences matter. The models are served by third parties at provider-default temperature, their decoding parameters are not under our control, and per-turn sampling seeds are not honoured, so their games are not paired with the open-weight games by common random numbers and are analysed separately throughout. The hosted interface returns only the final answer and no deliberation trace, so for these models we observe what was chosen and not what was considered. And they carry no comprehension probes.

(d) Group composition: the persona study

The baseline groups are homogeneous and neutral, which leaves target-reach at its ceiling in almost every cell and therefore unable to express a risk response. To break that ceiling we ran a second study that varies how prosocial the group is. Each agent's prompt was prefixed with a single sentence naming a disposition, either *selfish* ("You are a selfish player: you only care about maximising your own final amount of money") or *cooperative* ("You are a cooperative player: you want the whole group to reach the target and avoid the disaster"). We swept the number of selfish agents in the group from 0 to 6 in unit steps, giving seven compositions, and crossed this with the same three risk levels and two languages. Nothing else in the prompt changed; in particular the payoffs, the target and the stated catastrophe probability were identical across compositions, so composition carries no incentive information about the game itself. Persona-to-seat assignment was permuted deterministically from the condition and repetition indices. This scheme is not a uniform random assignment: it yields only four to nine distinct seat patterns per mixed composition, and across the mixed compositions the six seats are not occupied by a selfish agent equally often ($\chi^2 = 23.0$ on 5 d.f., $P = 0.0003$; two seats are never selfish when exactly one agent is), so we make no claims about seat position. We verified that this does not disturb the composition effect: adding the full seat arrangement to a model that already contains the number of selfish agents raises R^2 by only 0.005–0.029, against the 0.48–0.75 explained by the number alone. The composition study was run for the three smallest open-weight models (Qwen2.5-7B, Llama-3.1-8B, Gemma-2-9B) at ten repetitions, giving $1{,}260$ games, and for GPT-5.4-nano at five repetitions, giving 210 games.

(e) In-situ comprehension probes

To test whether behaviour was bounded by understanding, we adapted the meta-prompting battery of Fontana et al. [8] from the two-player prisoner's dilemma to the N-player collective-risk game. Twenty questions were organised along three axes: *Rules* (the fixed structure of the game, namely available actions, endowment, target, number of rounds and, critically, the catastrophe probability), *Time* (reading specific past events from the history) and *State* (the current cumulative situation, including the running pool total). Every question had a closed-form ground truth computed by the game engine, so grading involved no second model. Probes were issued on the identical prompts from which the agent was deciding and were appended after the decision, so a probe cannot influence the decision it accompanies. The comprehension runs were nevertheless a separate execution of the game, using greedy decoding for a deterministic benchmark, and their behavioural trajectories therefore differ from those of the main behavioural runs; the two studies characterise the same agents and the same prompts, not the same game trajectories. Because the shared pool total is the quantity most likely to expose a difference between reading and computing, we ran a controlled A/B manipulation in which the running total was either hidden (the default) or printed in the prompt. This manipulation adds no information about incentives, since the total is a deterministic function of a history the agent already has, so it serves as a salience control for the behavioural study as well as a comprehension probe. It was run for the five models in the comprehension panel at 7–32B, and yielded approximately $54{,}500$ graded probes per model. The comprehension panel used the same models as the behavioural study except that the 70B Llama was the Llama-3.3 revision rather than Llama-3.1; we keep the labels distinct throughout. Unparseable answers were graded as incorrect; the probe parse-failure rate was below 0.01% for six of the seven models and 1.03% for Llama-3.1-8B, where it was almost entirely confined to Vietnamese (2.05% of that model's Vietnamese probes). Accuracies are reported with Wilson 95% confidence intervals [16]. These treat probes as independent, which they are not, since probes are clustered within games, rounds and questions, so the intervals we plot are anti-conservative and should be read as a lower bound on uncertainty. Because the comprehension differences we rely on span tens of percentage points, this does not affect any conclusion drawn here, but it would matter for finer comparisons.

(f) Analysis and dependent variables

Our primary behavioural dependent variables were the target-reach rate (the fraction of games in which the pool met €120) and the group contribution (the pooled total, out of 240). We treated the realised catastrophe rate as secondary, because a single end-of-game lottery per game, with the draw shared across languages by the common-random-number scheme, yields only about ten effective draws per model and risk level and is therefore noisy. Contributions in the open-weight panel are analysed paired, because the design reuses sampling seeds across the risk and language conditions; the resulting within-pair correlations range from 0.32 to 0.95, so an unpaired analysis would discard real precision. Because reach is censored in most cells, we defined a subset of *uncensored* cells, those model-by-composition-by-language cells in which the reach rate is strictly between 0 and 1, so that the outcome is genuinely in doubt, and report the risk response within them as the study's decisive test for the open-weight panel (§3(a)). This criterion is a function of the outcome variable, so we report it alongside three checks that are not: selection on a disjoint risk level, selection on a pre-outcome criterion derived from the fitted composition line, and an unselected regression of contribution on risk, composition and their interaction over all composition games. We also report the minimum effect the design could have detected.

The commercial panel required a different analytical treatment, and it is worth saying why before the results rather than after. Those models are close to deterministic in this game: across the 36 configuration-by-language-by-risk cells the standard deviation of group contribution over ten games is below 3 points of 240 in 28 cells and exactly zero in 16. Within-model variance is therefore not the relevant source of uncertainty, and a within-model regression on risk is degenerate where a cell is constant. For the commercial panel we accordingly report effects as differences between cell means with unpaired Welch tests where variance permits, and we treat the informative contrasts as being *between* models: cheap tier against top tier within one family, and reasoning against non-reasoning within one base model. All quantities were recomputed directly from the raw per-game and per-probe logs.

3. Results

(a) The open-weight panel does not respond to risk, and the null survives where the outcome is in doubt

Humans in the collective-risk dilemma cooperate more when the catastrophe is more likely; the open-weight models did not. Every model reached the target at the same rate at every risk level (figure 2(a)), against a human benchmark that climbs from 0% to 50% across the same range. Six of the seven reached the target in essentially every game regardless of risk (Qwen2.5-7B, Llama-3.1-8B, Gemma-2-9B, Gemma-2-27B and Qwen2.5-72B at 100%; Qwen2.5-32B at 96.7%), and the seventh, Llama-3.1-70B, reached it in exactly half of its games at each of $p = 0.1$, 0.5 and 0.9.

That flatness is easy to describe and hard to test, and we are explicit about the limitation. Five of the seven models reached the target in all sixty of their games, so their reach rate has no variance and admits no regression on risk at all; a sixth varies only between 95% and 100%. Reach is thus censored at the ceiling for six of seven models, and a positive risk response of the human size would be arithmetically undetectable in it. We therefore treat reach as descriptive and take group contribution, which is free to move in either direction, as the informative dependent variable. Analysed paired, risk does move contributions. Every one of the seven models has a positive risk slope, and a paired t-test on the difference between $p = 0.9$ and $p = 0.1$ is significant for three of them: Gemma-2-9B (+8.8 points, $P = 0.0023$), Qwen2.5-72B (+3.2, $P = 0.0062$) and Gemma-2-27B (+1.9, $P = 0.048$); the first two survive Bonferroni correction across the seven models. The response is real. What it is not is consequential. The largest effect any model shows across the entire risk range is 8.8 of a possible 240, about one extra contribution of 2 from one agent in four of the sixty games, and it changes no outcome, since Gemma-2-9B already reached the target in every game at every risk level. Against the human benchmark, where the identical manipulation moves the fraction of successful groups from none to half, this is a response in the right direction and of the wrong order of magnitude. Gemma-2-9B's response also deserves the caveat we give Llama-3.1-70B below: it is entirely Vietnamese, since in English the model contributed exactly 120.0 in all thirty games, a hard constant with no variance and no slope to estimate, while in Vietnamese it rose from 127.8 to 145.4.

The censoring objection is the strongest available defence of these models, and it deserves a direct test rather than a caveat. If reach sits at its ceiling, a risk response cannot appear in it, and the flat curves of figure 2(a) would be uninformative about whether the agents are risk-sensitive. The composition study supplies the missing conditions. Varying the number of selfish agents moves groups off the ceiling and, in some cells, onto the knife edge, where roughly half of the games succeed and half fail. Those are the conditions in which a risk-responsive agent has most to gain from contributing more, and in which the outcome variable is free to move in either direction. We therefore defined the uncensored cells as those with a reach rate strictly between 0 and 1, and tested the risk response inside them. Seven of the 42 open-weight cells qualify and three of the 14 GPT-5.4-nano cells. The result is a clean null. Pooling the open-weight uncensored cells and pairing on the sampling seed, raising the catastrophe risk from $p = 0.1$ to $p = 0.9$ changes group contribution by +0.97

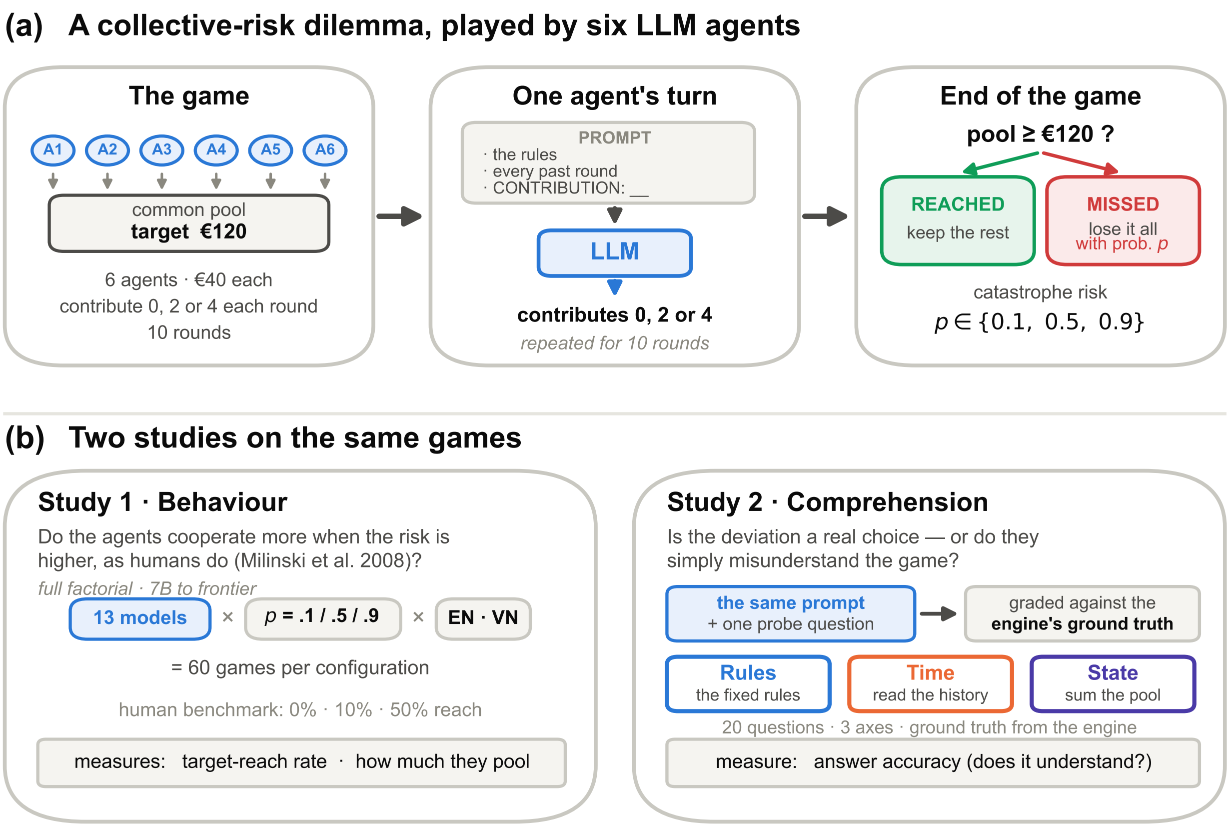

Figure 1. Study design. (a) Six LLM agents play a faithful replication of the Milinski collective-risk dilemma: each of six agents starts with €40 and, over ten rounds, privately contributes 0, 2 or 4 to a common pool from a prompt containing the rules and the full history; at the end, if the pool has not reached €120 the group suffers a catastrophe with probability $p \in \{0.1, 0.5, 0.9\}$. (b) The same games feed two studies. Study 1 (behaviour) crosses the model panel with three risk levels and two languages and measures the target-reach rate and how much the agents pool. Study 2 (comprehension) appends a graded question to the identical prompt, scoring twenty questions along three axes (Rules, Time, State) against the engine's ground truth, to ask whether any deviation is a genuine choice or a misunderstanding.

points of 240 (95% CI $[-2.2, +4.1]$, $n = 70$ pairs, $P = 0.54$), and the reach rate by $+4.3$ percentage points ($0.514 \to 0.557$). In GPT-5.4-nano the point estimate is larger but no more distinguishable from zero: $+7.5$ points (95% CI $[-6.3, +21.2]$, $n = 15$ pairs, $P = 0.26$), with reach moving $0.400 \to 0.467$. Not one of the ten individual uncensored cells shows a significant risk effect at $\alpha = 0.05$, and four of the ten point in the wrong direction.

This null is not a failure of power. The paired differences in the open-weight uncensored set have a standard deviation of 13.1 points, so with 70 pairs the design detects an effect of 4.4 points of 240 with 80% power; the observed estimate is $+0.97$ and its upper 95% bound is $+4.05$. We can exclude, at conventional confidence, anything larger than a 4-point response, against a human benchmark in which the same manipulation moves half of all groups across the threshold, a shift of 120 points in the groups it moves. Selecting cells on a function of the outcome variable is the obvious objection to this analysis, so we checked it three ways, none of which selects on the tested data. First, choosing the cells using only the $p = 0.5$ games and testing on the disjoint $p = 0.1$ and $p = 0.9$ games returns the same estimate ($+0.97$, $P = 0.54$). Second, selecting cells by a criterion computed before any outcome is observed, namely those whose composition places the fitted contribution line within ± 30 points of the threshold, widens the set to 13 cells and 130 pairs and gives $+1.34$ points (95% CI $[-0.7, +3.4]$, $P = 0.19$). Third, abandoning selection entirely and fitting the risk-by-composition interaction to all 1,260 open-weight composition games gives a risk coefficient of $+3.80$ per unit probability (95% CI $[-3.5, +11.2]$, $P = 0.31$), implying $+3.0$ points across the full $p = 0.1 \to 0.9$ sweep, with no significant interaction ($P = 0.21$); the corresponding GPT-5.4-nano estimate is -0.07 points ($P = 0.99$). Every route to the question returns the same answer. In precisely the conditions where the ceiling argument does not apply, where the group is balanced on the threshold and the catastrophe is a live possibility rather than a formality, a ninefold change in the probability of losing everything moves behaviour by an amount indistinguishable from zero.

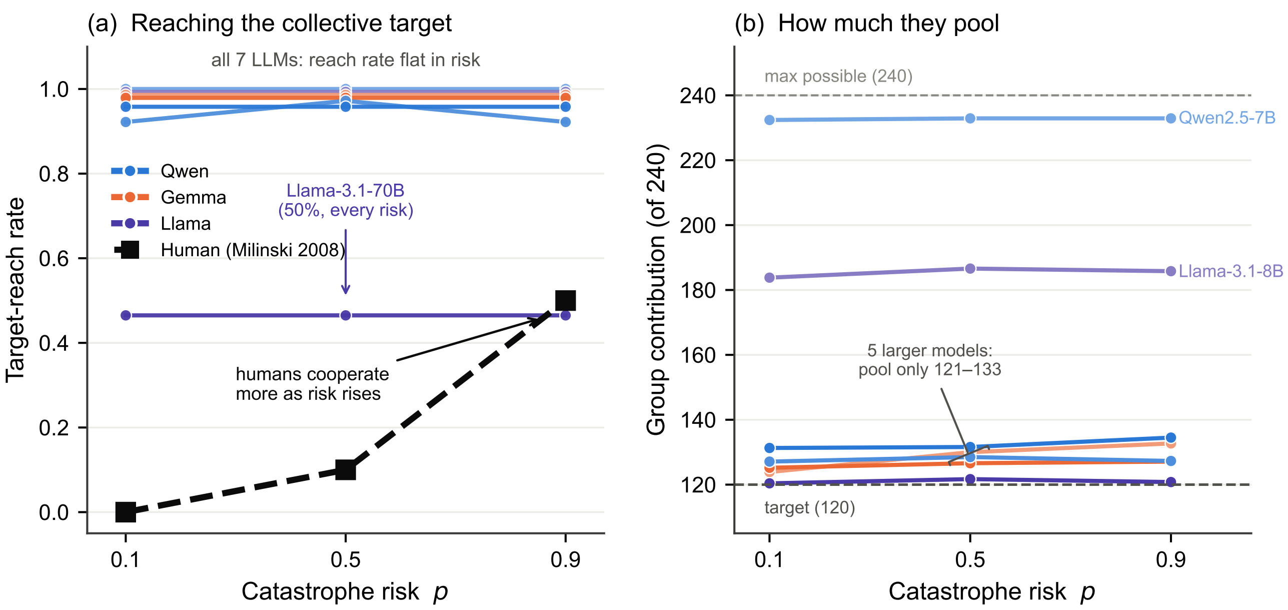

Figure 2. No open-weight model reproduces the human risk response. (a) Target-reach rate against catastrophe risk for all seven open-weight models (colour denotes family; shade denotes size) and for humans (dashed). Every model's reach rate is flat in p; humans rise from 0% to 50%. Llama-3.1-70B is flat at 50%. (b) Group contribution (of a possible 240) against risk. Contributions are essentially flat in p (the one weak exception, Gemma-2-9B, is quantified in the text); the vertical spread reflects how economically each model plays, not risk sensitivity. In (a) five models sit exactly on 100%, so the curves are drawn with a small downward offset (at most 0.04) purely so that all seven remain visible; no curve is drawn above its true value.

Two further regularities in this panel are worth recording, because both will turn out to have counterparts at the frontier. The first is that scale changes how much the models cooperate without changing whether they attend to risk. In every family the largest model pooled far less than the smallest, though the decline is not strictly monotonic (figure 3): Qwen2.5 fell from 232.7 of a possible 240 at 7B to 127.6 at 32B before ticking back up to 132.5 at 72B, Llama fell from 185.4 at 8B to 121.0 at 70B, and Gemma was already economical at 9B (128.9) and stayed there at 27B (126.3). The panel separates into a ceiling-hugging Qwen2.5-7B, an intermediate Llama-3.1-8B, and a cluster of five larger models that contribute just enough to clear €120 and keep the rest. This economy is orthogonal to risk: the more economical models are not the more risk-sensitive ones, they are simply better at extracting private surplus while still clearing the collective target. The second regularity is that the one model that fails does so by language rather than by risk. Llama-3.1-70B's pooled average of 121.0 appears to place it exactly on the target, as though it were playing a knife-edge strategy, but that average is a mixture of two well-separated modes: in English it pooled 130.3, with every game between 124 and 140, and reached the target in all thirty games, while in Vietnamese it pooled 111.6, with every game between 108 and 118, and reached the target in none. Only five of its sixty games fall anywhere in the interval 115–125. Because its Vietnamese games always missed, this model absorbed 22 of the 23 catastrophes suffered in the baseline study, and the behaviourally meaningful point is that the punishment changed nothing: within each language its contribution was flat across risk, so the one model that was repeatedly ruined did not respond by cooperating more.

(b) What the prompt does that the incentive does not

If a ninefold change in the probability of ruin barely moves these agents, what does? Three manipulations do, and none of them carries any information about payoffs, target or risk. They are the reason we describe the risk null as informative rather than merely negative: they are measured on the same games, under the same decoding, with the same dependent variable, so the comparison of magnitudes is internal.

The first is pure salience. Printing the running pool total in the prompt adds nothing an agent did not already have, since the total is a deterministic function of a history it is shown in full. It nevertheless moves behaviour by an order of magnitude more than risk does (figure 4). Contributions fell by 95.3 points for Qwen2.5-7B ($232.5 \to 137.2$, paired $P = 6 \times 10^{-34}$), by 26.6 for Llama-3.1-8B ($P = 2 \times 10^{-17}$), by 8.2 for Gemma-2-9B ($P = 2 \times 10^{-6}$) and by 2.9 for Qwen2.5-32B ($P = 0.003$). The fifth model, Gemma-2-27B, shifted by the same 2.9 points but not significantly ($P = 0.16$); its target-reach rate nevertheless fell from 100% to 67%, because it plays so close to the threshold that a shift too small to detect in contributions is large

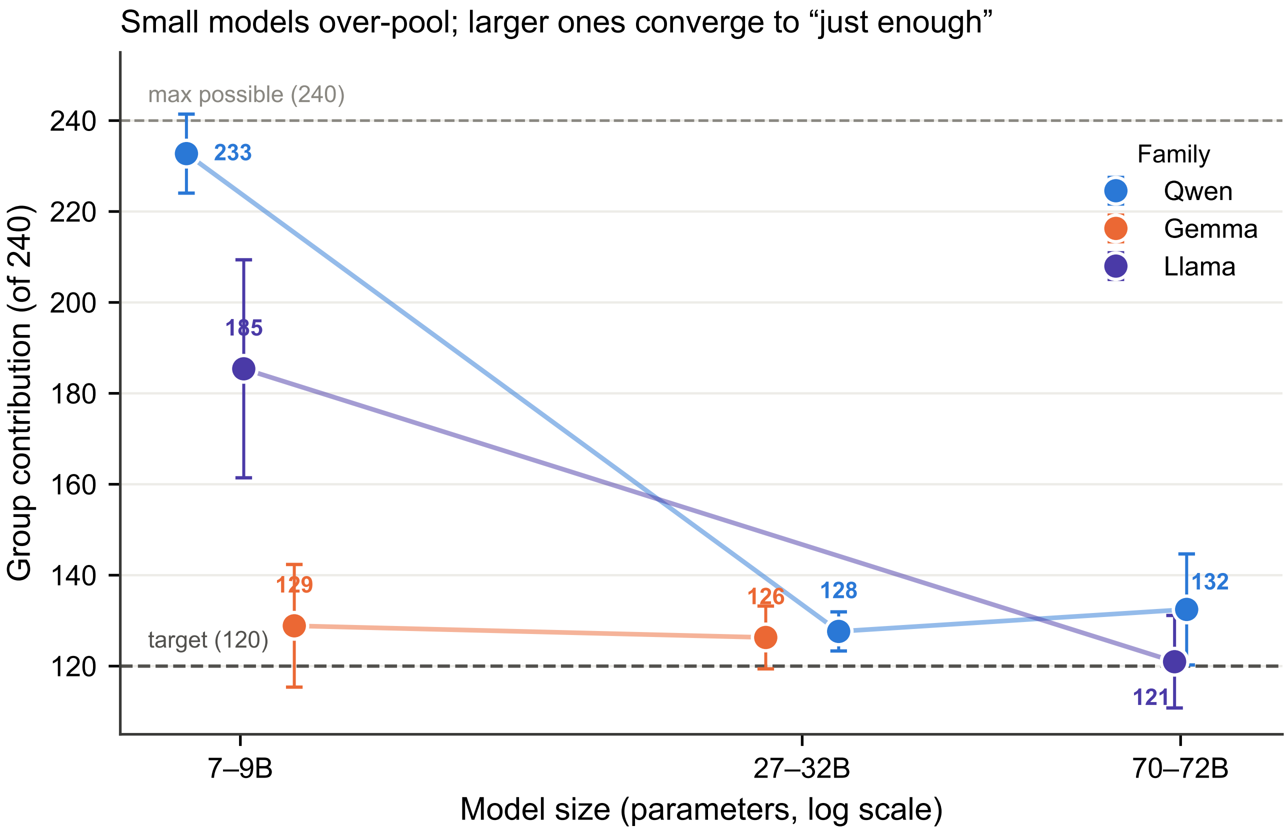

Figure 3. Small models over-pool; larger ones converge to "just enough". Group contribution (of 240) against model size on a logarithmic axis, connected within family. In every family the largest model pools far less than the smallest, but the decline is not strictly monotonic: Qwen2.5 falls to 127.6 at 32B and rises again to 132.5 at 72B. Error bars are standard deviations across games.

enough to change the outcome. The two smallest models, which over-contributed most when the total was hidden, are precisely the ones that correct hardest when it is shown.

The second is who is in the group. Replacing cooperative with selfish personas one agent at a time moves contributions monotonically and by a large margin: from all-cooperative to all-selfish, group contribution falls by 100.4 points for Gemma-2-9B ($148.1 \rightarrow 47.7$, paired $P = 6 \times 10^{-37}$), by 94.0 for Llama-3.1-8B ($199.4 \rightarrow 105.4$, $P = 3 \times 10^{-24}$), by 31.8 for Qwen2.5-7B ($237.7 \rightarrow 205.9$, $P = 4 \times 10^{-28}$) and by 112.4 for GPT-5.4-nano in English. Even the smallest of these is more than three times the largest risk effect in the open-weight panel, and the largest is more than ten times it. What makes this manipulation informative is not only its size but its regularity. Group contribution falls almost perfectly linearly in the number of selfish agents (figure 5(a)), with within-language R^2 of 0.95 for both Gemma-2-9B and Llama-3.1-8B in English, 0.63–0.85 elsewhere, and 0.89 and 0.88 in the two languages for GPT-5.4-nano. Each model is therefore summarised by two numbers, an intercept, the contribution of an all-cooperative group, and a slope, the cost of converting one cooperative agent into a selfish one: pooled across languages these are 146.5 and -16.6 for Gemma-2-9B, 199.4 and -15.9 for Llama-3.1-8B, 234.9 and -5.0 for Qwen2.5-7B, and 210.5 and -18.8 for GPT-5.4-nano in English.

That two-parameter description explains something that looks, on the raw reach curves, like three qualitatively different failure modes (figure 5(b)). Gemma-2-9B's reach rate collapses immediately, halving at a single selfish agent and reaching zero by five. Llama-3.1-8B holds at or near 100% through four selfish agents and then falls off a cliff, to 52% at five and 28% at six. Qwen2.5-7B never fails at all: it reaches the target in every one of its 420 composition games, even when all six agents are told to be selfish. Yet all three obey the same linear law, and the difference is entirely where each line starts and how steeply it falls relative to a threshold that does not move. Gemma's line crosses 120 at 1.6 selfish agents, Llama's at 5.0, GPT-5.4-nano's at 4.8, and Qwen's only at 23.0, that is, never, since the group has six members. Dramatic differences in apparent robustness are the arithmetic consequence of a linear response meeting a fixed threshold, not evidence of different strategies. It is also in these games that the decisive risk test of §3(a) lives, and it is worth noting what the balanced cell shows directly: at three selfish agents against three cooperative ones, where groups are furthest from any ceiling, reach remains flat in risk for all three open-weight models and group contribution varies by less than 2.2 points across the entire risk range. Making the group's fate genuinely uncertain does not make these agents attend to the probability that decides it.

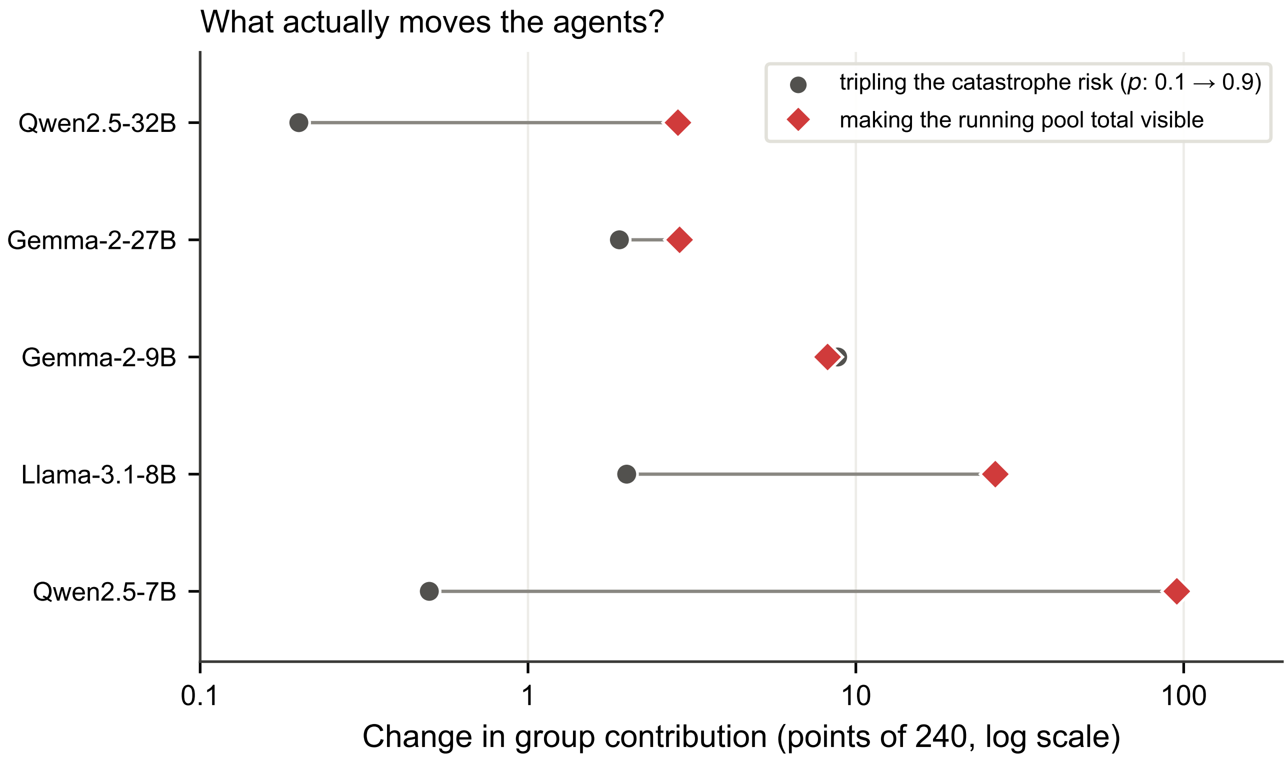

Figure 4. Salience beats incentive. Absolute change in group contribution (of 240, log axis) produced by two manipulations, for the five models run in both arms: tripling the catastrophe risk from $p = 0.1$ to $p = 0.9$ (grey circles, paired within seed) versus printing the running pool total in the prompt (red diamonds). The salience manipulation carries no information about payoffs, target or risk, yet dominates for four of the five models.

The third is the language the prompt is written in, and in one model it is the largest single effect we measured anywhere. GPT-5.4-nano pooled 209.7 in English and 62.9 in Vietnamese, a paired difference of 146.8 points of 240 ($P = 1 \times 10^{-16}$, $n = 15$), and its target-reach rate went from 100% to 0%. This is not a marginal shift in a model living near the threshold, as it was for Llama-3.1-70B: the Vietnamese groups pooled roughly half of the target, and 84 of the 105 Vietnamese composition games ended in a realised catastrophe. In Vietnamese the model begins below the threshold even when all six agents are told to be cooperative (78.7 of 240), so the composition line never crosses 120 at any composition and reach is floored at zero throughout (figure 6(b)), the mirror image of the ceiling censoring that limits the English cells. Placed on a common axis (figure 6(c)), the four manipulations separate by orders of magnitude for this model: catastrophe risk moves group contribution by 4.4–7.2 points, group composition by 65.1–112.4, and language by 146.8. The one manipulation the game says should matter is the smallest of the four by a factor of nine to thirty-three.

(c) At the top of the frontier the null breaks, but only for two models

Everything to this point describes a panel that ignores the quantity the game is built around. Extending the panel to the top commercial tier does not sharpen that conclusion; it breaks it, and the break is the central result of the study.

Two of the six commercial models play the expected-value solution, and they play it almost exactly (figure 7(a)). Gemini-3.1-Pro contributed nothing whatsoever at $p = 0.1$, a group total of 0.0 of 240 in all ten English games with zero variance, and then contributed exactly 120.0, the target and not one unit more, in all twenty games at $p = 0.5$ and $p = 0.9$. GPT-5.6-sol did the same, pooling 1.6 at $p = 0.1$, 120.0 at $p = 0.5$ and 119.8 at $p = 0.9$. The English risk effect is $+120.0$ points of 240 for Gemini-3.1-Pro, a difference between two constants for which no test is defined, and $+118.2$ for GPT-5.6-sol ($P = 1 \times 10^{-22}$), roughly fourteen times the largest effect anywhere in the open-weight panel. Both hold up in Vietnamese, at $+119.6$ and $+74.2$ respectively.

These agents are not merely moving; they are winning. At $p = 0.1$ an agent that contributes the target share ends with 20 for certain, whereas one that withholds everything keeps 40 and faces a one-in-ten lottery. Gemini-3.1-Pro's agents realised 32.0 each and GPT-5.6-sol's 31.8, against 20.0 for Claude-Opus-5 and for Grok-4.20 with reasoning, the two top-tier configurations that paid exactly the target at that risk level, and less still for those that over-contributed (19.7 for Gemini-3.1-Flash-Lite, 7.1 for Grok-4.20 without reasoning, 5.7 for GPT-5.4-nano). The two models that walked away therefore earned 60% more than the best-paid cooperator. The realised figures are below the expected 36 because two of the ten

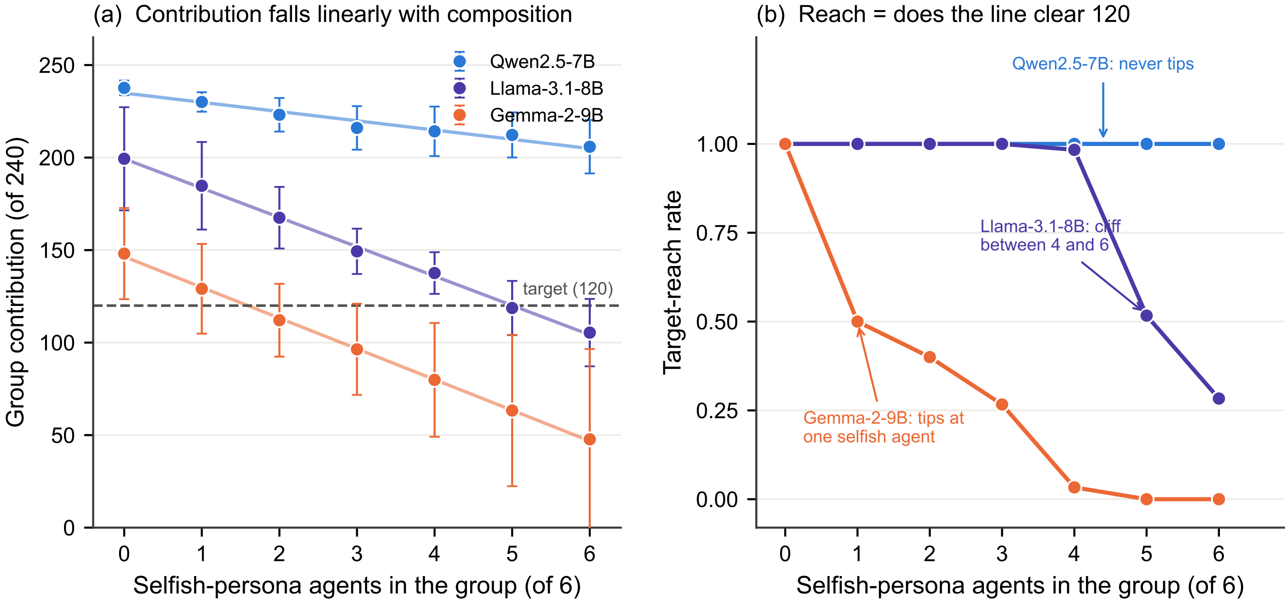

Figure 5. Group composition is a linear engine against a fixed threshold. (a) Group contribution (of 240) against the number of selfish-persona agents in the six-agent group, for the three open-weight models run in this study; lines are ordinary least-squares fits, points are cell means and error bars are standard deviations across games. The response is close to linear in every model (R^2 up to 0.95 within language). (b) The corresponding target-reach rates. The three apparently distinct failure modes, immediate collapse, a late cliff, and no failure at all, follow from where each model's line in (a) crosses the fixed target of 120: at 1.6, 5.0 and 23.0 selfish agents respectively.

shared lottery draws fell under 0.1 and cost those groups everything, which is exactly what a risk-neutral agent accepts in exchange for the 90% of games in which it keeps the full endowment. The behaviour is not merely thrift, either. At $p = 0.5$, the point at which the two options have identical expected value, both models switch to cooperating and do so at exactly the minimum, contributing 120.0; and at $p = 0.9$ they cooperate again, GPT-5.6-sol missing the target by a single contribution in one of ten games (118 of 120, giving a 90% reach rate). The pattern across the three risk levels tracks the sign of the expected-value comparison, not a preference for either withholding or contributing.

The other four configurations do not. Claude-Opus-5 reached the target in all sixty of its games, with cell means spanning 120.0 to 128.0; its English risk effect of $+8.0$ points is highly significant ($P = 3 \times 10^{-6}$) and behaviourally irrelevant, being 3% of the board and never altering an outcome, and it falls to $+1.2$ ($P = 0.41$) in Vietnamese. Grok-4.20 in its reasoning configuration is the most inert model in the entire study, contributing 120.0 in twenty-nine of its thirty English games and 120.0 in all thirty Vietnamese ones, for a risk effect of $+0.2$ and $+0.0$. Grok-4.20 with reasoning disabled contributes far more than it needs to, 197.2 at $p = 0.1$ rising to 220.8 at $p = 0.9$, and its $+23.6$-point slope is the largest at the top tier outside the two expected-value models, but it is not significant ($P = 0.098$) and, more tellingly, it is a rise in the amount of over-contribution rather than a movement across any decision boundary: the model reaches the target in every game at every risk level, including the level at which reaching it is the wrong thing to do. At the cheap tier, Gemini-3.1-Flash-Lite moved -1.8 points, in the wrong direction, and GPT-5.4-nano $+7.5$.

The consequence for how such results should be read is visible in the reach rates (figure 7(b)). Target-reach at $p = 0.1$ in English divides the panel perfectly and with nothing in between: 0% for the two expected-value models and 100% for every other configuration in the commercial panel. Had we run this study on any one top-tier model, as cost pressures normally dictate, we would have drawn a confident conclusion, and with probability one half it would have been the opposite of the truth.

(d) Capability is necessary but not sufficient, and reasoning buys thrift

What distinguishes the two models that solve the game from the eleven that do not? Not the provider: the four top-tier models come from four different laboratories and divide two against two, so no vendor's alignment or training recipe accounts for the split. Not the presence of explicit deliberation: all four top-tier models are reasoning systems. And not capability alone, although capability is clearly involved. Figure 7(c) places all fourteen configurations on one axis, and the ordering is neither by size nor by tier nor by vendor. It is, as far as our design can tell, a property of the individual model.

Two contrasts within the panel isolate what capability does and does not buy, and both are internal, changing one factor while holding the rest fixed. The first is the capability contrast, and it is the cleanest in the study because it holds

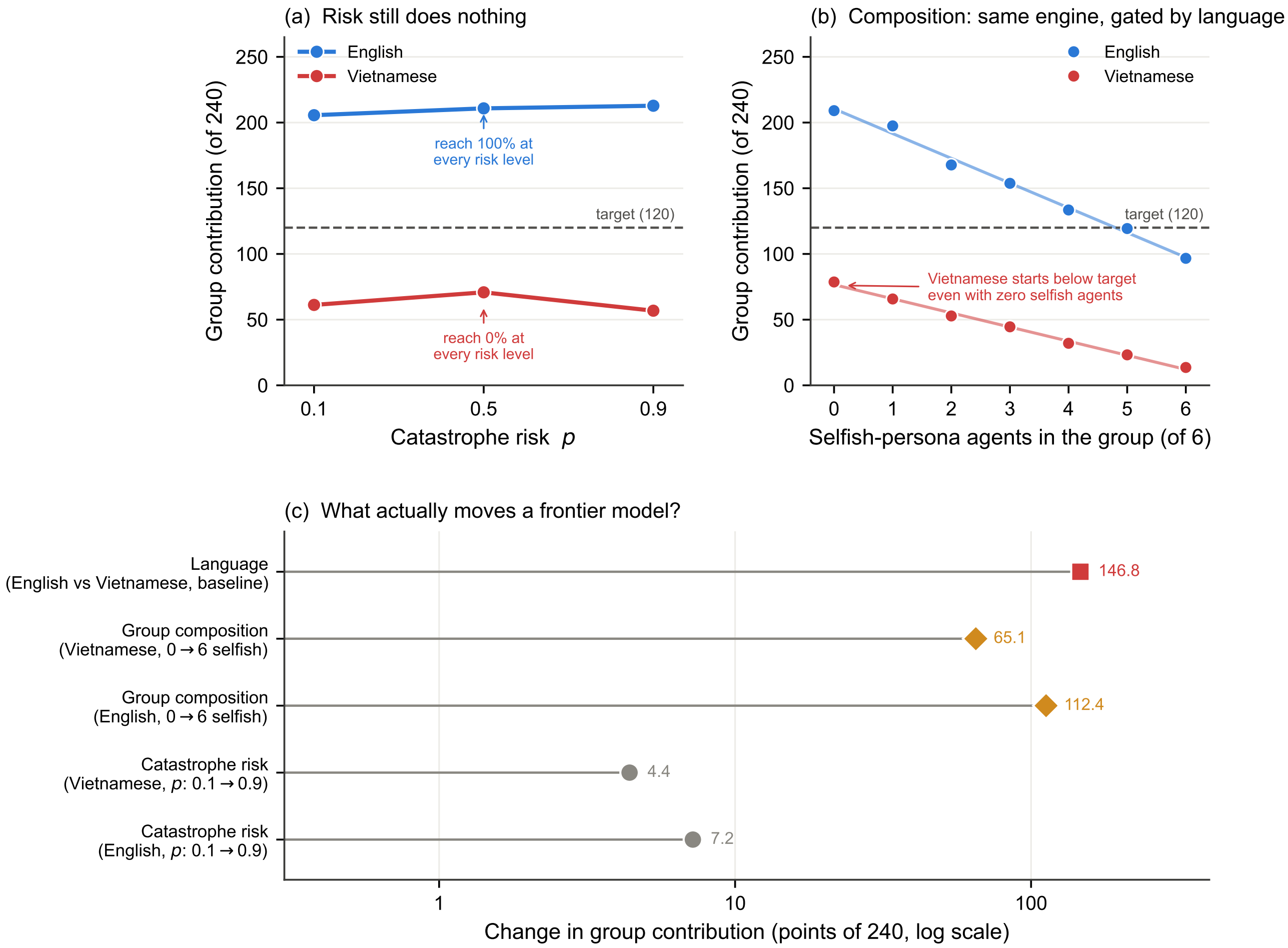

Figure 6. A cheap frontier model shows the open-weight pattern, and an extreme language effect. GPT-5.4-nano, 30 baseline and 210 composition games. (a) Group contribution against catastrophe risk, by language: flat in both, with English reaching the target in 100% of games at every risk level and Vietnamese in 0%. (b) The composition study shows the same near-linear engine as the open-weight models ($R^2 = 0.89$ and 0.88 in the two languages), but the Vietnamese line starts below the target even with zero selfish agents, so it never crosses. (c) The four manipulations on a common logarithmic axis. Risk is the smallest by a factor of nine to thirty-three.

the provider, the generation and the interface constant: Gemini-3.1-Flash-Lite and Gemini-3.1-Pro are the cheap and top tiers of one family, and their risk effects are -1.8 and $+120.0$ points respectively. The cheap model reaches the target in every game at every risk level; the expensive one refuses to at $p = 0.1$ and then contributes exactly the target thereafter. Together with the observation that no model below the top commercial tier anywhere in our panel, open-weight or cheap-tier commercial, ever plays the expected-value solution, this supports a two-part claim that is stronger than either half alone: capability is necessary, since risk-responsive play appears only at the top tier, and capability is not sufficient, since half of the top tier does not exhibit it. The claim also answers the objection that most naturally meets a null result of this kind, namely that the models tested were simply not good enough.

The second contrast isolates explicit reasoning by toggling it within one base model. With reasoning disabled, Grok-4.20 pools 197.2 at $p = 0.1$ and 220.8 at $p = 0.9$ in English, and saturates in Vietnamese at 239.4 of a possible 240, tipping essentially the entire endowment into the pool every round; with reasoning enabled, the same model contributes 120.0 at every risk level in both languages. Enabling reasoning is therefore worth 77.2 points of 240 at $p = 0.1$ in English, and 105.1 points pooled over all six cells, converting a model that gives away its endowment into one that pays the exact price of the target and keeps the rest. It is worth nothing at all in risk sensitivity: the risk effect falls from $+23.6$ to $+0.2$, and at $p = 0.1$ the reasoning configuration still contributes 120 where the expected-value solution contributes 0. Deliberation, in this model, is spent on the question of how cheaply the target can be reached and not on the prior question of whether it is worth reaching. That is a sharper statement of the decoupling between capability and incentive-responsiveness than scale alone provided, because it is a within-model manipulation of the very faculty one would expect to produce the missing behaviour.

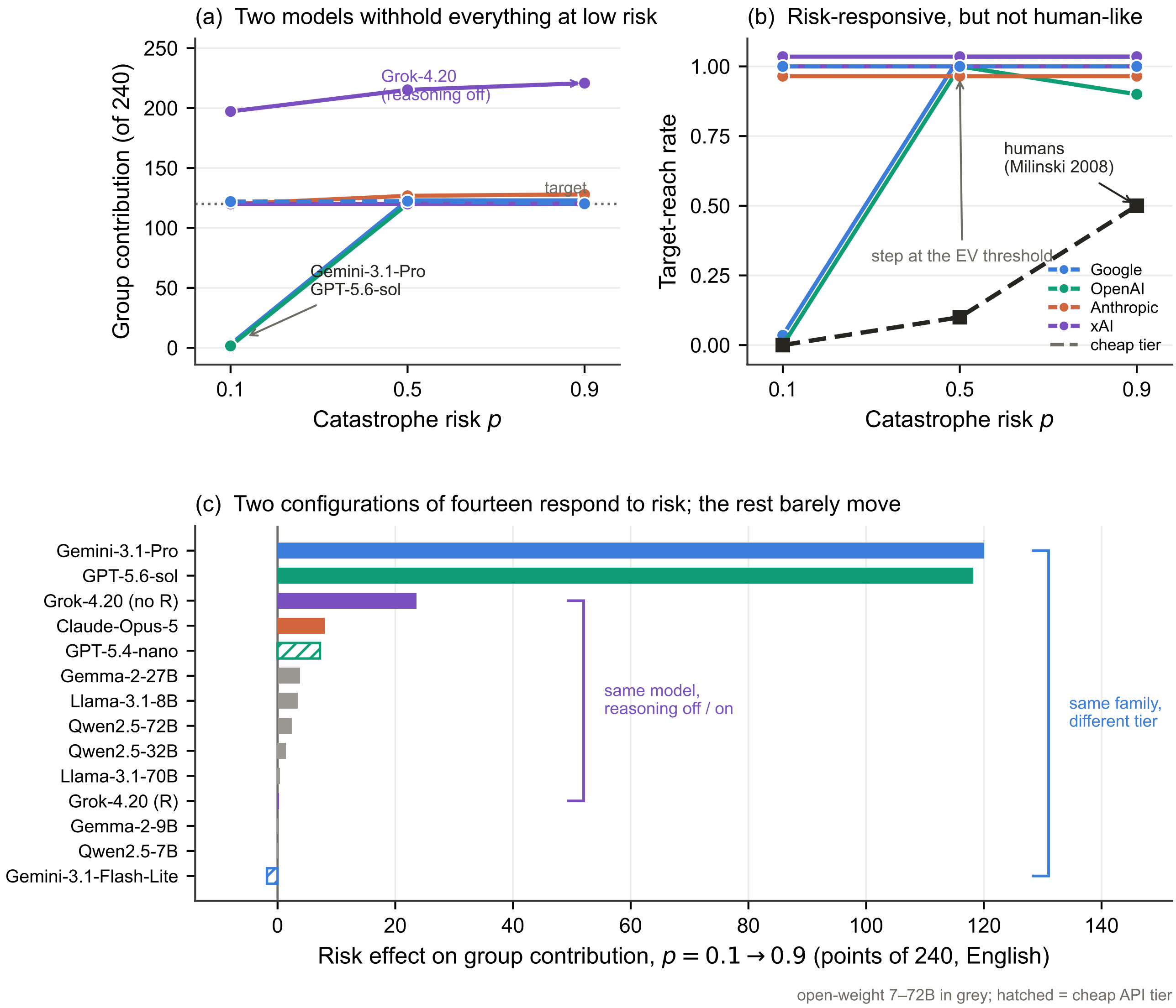

Figure 7. Risk sensitivity is a property of the model, not of the tier. (a) Group contribution (of 240) against catastrophe risk in English for the six commercial models; solid lines are the top capability tier, dashed the cheap tier, colour denotes provider. Gemini-3.1-Pro and GPT-5.6-sol withhold everything at $p = 0.1$ and then contribute exactly the target; the remaining configurations are flat. (b) The corresponding target-reach rates against the human benchmark (black, dashed). The two responsive models form a step function at the expected-value threshold, more decisive than the graded human curve rather than blinder than it. (c) Risk effect on group contribution across all fourteen configurations of the thirteen-model panel, sorted; open-weight models in grey, cheap commercial tiers hatched. The two bracketed pairs are the study's controlled contrasts: the same family at two capability tiers, and the same base model with explicit reasoning disabled and enabled. In (a) Gemini-3.1-Pro is drawn 3 points high and in (b) three configurations are offset by at most 0.035, purely so that coincident curves remain visible; the exact values are given in table 1.

Finally, the risk-responsive models are not thereby human-like, and the difference matters for anyone proposing to use LLM agents as surrogates. Human groups in this dilemma reach the target in 0%, 10% and 50% of games as risk rises, a graded response that leaves most groups failing even when the catastrophe is nearly certain. Gemini-3.1-Pro and GPT-5.6-sol go 0%, 100%, 100% (figure 7(b)). They are a step function at the expected-value threshold, agreeing with humans exactly where the expected-value solution and human behaviour happen to coincide, at $p = 0.1$, and departing from them sharply everywhere else. The deviation from the human benchmark at the top tier is therefore an excess of decisiveness rather than a deficit of attention, which is the opposite failure mode from the one displayed by the other eleven models. Language shifts these agents too, and in model-specific directions that resist aggregation: Vietnamese leaves Gemini-3.1-Pro untouched (+119.6 against +120.0), degrades GPT-5.6-sol's expected-value play from +118.2 to +74.2 and inflates its standard deviation at $p = 0.1$ from 1.6 to 47.7 as three of its ten Vietnamese groups revert to near-cooperative play, pooling 114 and falling just short of the target, saturates Grok-4.20 without reasoning at the ceiling, and, in GPT-5.4-nano, inverts the outcome entirely. Four models, four directions: there is no single language effect to report.

(e) What the agents can and cannot do

The behavioural deviation of the eleven risk-insensitive models might, in principle, be a failure of comprehension: perhaps they over-cooperated and ignored the risk because they did not grasp the rules or could not read the game. The in-situ probes rule this out for the open-weight panel. Accuracy on the *Rules* axis was uniformly high, from 93.8% for Llama-3.1-8B to 100% for the five largest models, and accuracy on the *Time* axis, reading specific events from the history, ranged from 82% to 99% (figure 8). One probe asked the model to state the catastrophe probability of the game it was playing, and every model answered almost always correctly: 98.1% for Qwen2.5-7B, 91.4% for Llama-3.1-8B and 100% for the other five (figure 9(b), upper curve). We are deliberately modest about what this shows. The decision prompt prints the catastrophe probability verbatim, so the probe is a retrieval task, and these models copy printed values at ceiling throughout our data. It establishes that the risk was legible in the context the agent decided from, which is worth establishing since it rules out the possibility that the number was lost or garbled, but it does not show that the agent represented the risk as a decision-relevant quantity, and it cannot distinguish an agent that weighed the risk and set it aside from one that never entered it into a calculation at all.

The only axis on which the models struggled was *State*, the current cumulative situation, and the reason is specific and revealing. When we asked for a quantity that was printed verbatim in the prompt, for example a player's own remaining money, accuracy was high for every model: under the same hidden-pool prompt used for the comparison below, five of the seven answered at or within a fraction of a point of ceiling, and Qwen2.5-7B scored 89.0%. The exception is Llama-3.1-8B, at 68.8% overall and only 46.0% in English; that model cannot reliably read the printed value either, so it is a poor test of the dissociation, and the contrast below should be read from the other six. When we asked for the same kind of quantity that had to be *summed* across rounds, namely the running pool total while it was hidden from the prompt, small models collapsed: Qwen2.5-7B answered correctly only 14.3% of the time and Llama-3.1-8B only 9.8%, even though both could read a printed value almost perfectly (figure 9(a)). That the failure is arithmetic rather than perceptual is confirmed by the A/B manipulation: when the same pool total was printed rather than hidden, accuracy jumped to 92–100% across the board. Reading, in other words, is not adding. This cumulative-addition deficit is a property of small models that scale largely removes. Accuracy on the hidden pool total rose steeply, if not perfectly monotonically, with size: 14.3% and 9.8% for the 7B and 8B models, 21.2%, 61.5% and 43.5% for the 9B, 27B and 32B models, and 81.7% and 88.2% for the 70B and 72B models (figure 9(b), lower curve); the two local reversals fall between families rather than within them. The juxtaposition in figure 9(b) is the one that matters: knowledge of the risk is flat and near-perfect across the whole size range, while the ability to track cumulative state climbs steeply with scale. By 70–72B the models both know the risk and can add the pool, and they still ignore the risk.

Language separates these two capacities cleanly (figure 10). Behaviourally, the two languages produced identical target-reach rates for five of the seven open-weight models; Qwen2.5-32B dipped slightly (100% in English versus 93.3% in Vietnamese) and Llama-3.1-70B collapsed outright, from 100% to 0%. In comprehension, language left rule and risk knowledge essentially untouched, with Rules accuracy differing by under one percentage point between languages and the risk probability read correctly in both, but it selectively attacked the arithmetic: State accuracy fell from 73.6% in English to 61.8% in Vietnamese, an almost twelve-point drop. The damage was concentrated in the middle of the range rather than at the bottom, with Gemma-2-9B losing the most (87.8% to 56.4%, 31.5 points) while the two models that were already worst at cumulative addition had the least left to lose (Qwen2.5-7B 13.6 points, Llama-3.1-8B 11.3 points) and Llama-3.3-70B was unaffected (Vietnamese in fact 0.8 points higher). Language, then, changes what these models can compute far more than what they know. We emphasise that this gate is available only for the open-weight panel: the commercial models, including the two that solve the game, carry no comprehension data, so we can say that the eleven risk-insensitive open-weight and cheap models did not misunderstand the game, and we cannot say by what internal route the two responsive models arrive at the expected-value solution.

4. Discussion

The central result of this study is a dissociation measured under one game. Across seven open-weight models spanning an order of magnitude of scale, two cheap commercial models and two of the four top-tier ones, a ninefold increase in the probability of losing everything moves cooperation by no consequential amount, and by an amount statistically indistinguishable from zero wherever the outcome was free to move, while three manipulations that carry no information about payoffs, target or risk each move it by tens to over a hundred points: displaying a number the agent already had the information to compute, changing which dispositions are named in the six prompts, and changing the language those prompts are written in. The unifying observation for those eleven models is that all three manipulations that work are delivered through the prompt, and the one that does not work is delivered through the incentive structure. It is not that these agents fail to respond to their situation, since they respond vigorously, and in the normatively sensible direction, to being told who they are and to being shown the state of play. It is that the channel through which a game communicates stakes, namely the consequences attached to outcomes, is the channel they attend to least. An agent that adjusts its

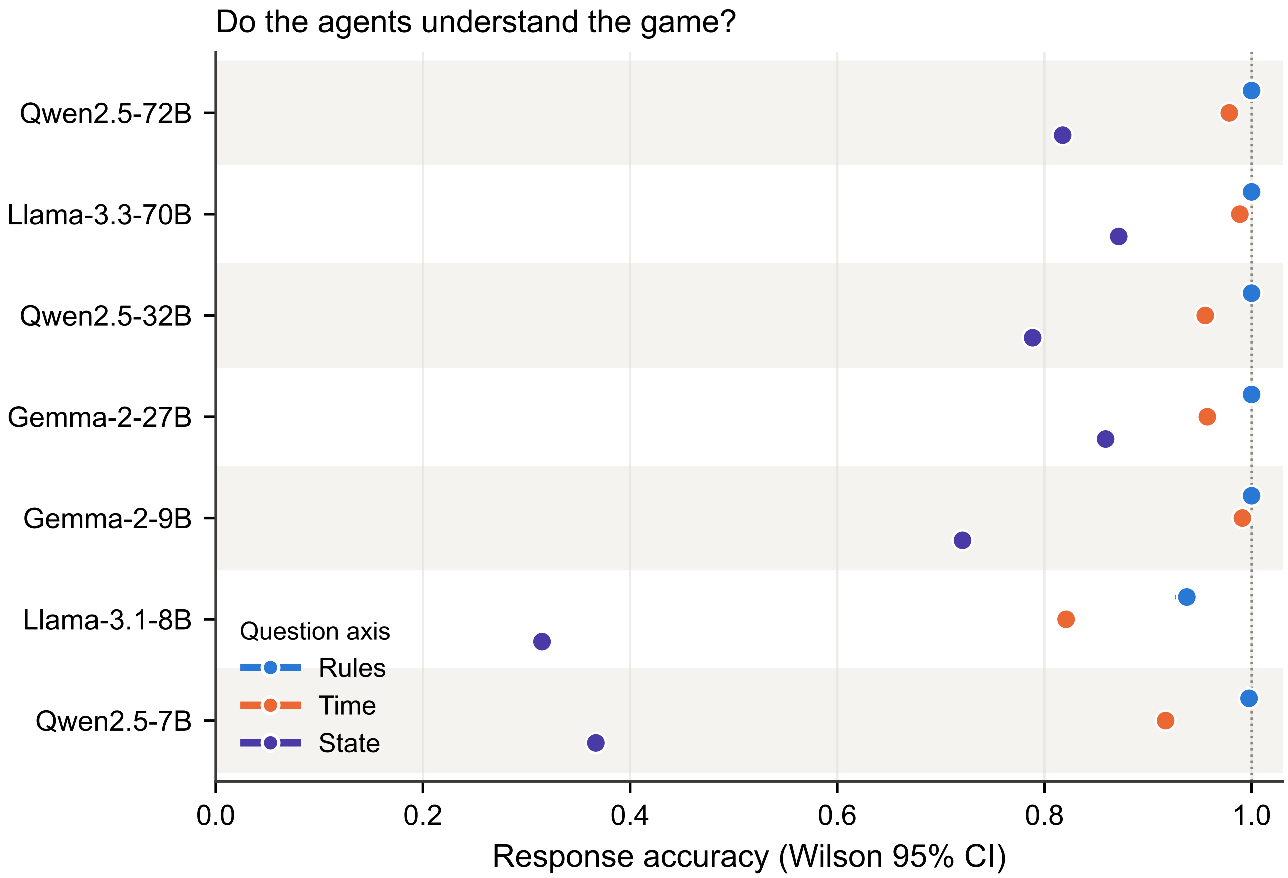

Figure 8. Comprehension by question axis. Response accuracy (dots) with Wilson 95% confidence intervals for each model, ordered by size (smallest at the bottom), on the Rules, Time and State axes; alternating bands group the three dots belonging to one model, and with $2{,}880$–$28{,}200$ probes per point the intervals are narrower than the markers. Rules is at ceiling and Time is high (82–99%) for all models; State (cumulative situation) is the only axis that varies widely, and it improves broadly with scale.

contribution by a hundred points when a neighbouring prompt says “selfish”, and by one point when its own probability of losing everything rises from one-in-ten to nine-in-ten, is not weighing costs and benefits in any recognisable sense.

The other two models weigh them precisely. Gemini-3.1-Pro and GPT-5.6-sol give away nothing when the catastrophe is unlikely, pay exactly the target when it is not, and are paid 60% more for the difference. This is not a small quantitative improvement on the panel; it is a categorical change in what governs the behaviour, and it is what converts our null from a claim about language models into a claim about most language models. Any evaluation of this kind that had been run on one model, as budget normally dictates, would have reported one of two opposite conclusions with equal confidence, and our own earlier draft, built on a single cheap frontier model, reported the wrong one. The methodological lesson generalises well beyond this game: where a behaviour is a property of individual systems rather than of the technology, the sample size that matters is the number of models, and it is the axis on which behavioural evaluations of LLMs are most systematically underpowered.

What the split is not is as informative as what it is. It is not a provider effect, since four top-tier models from four laboratories divide two against two. It is not the presence of explicit reasoning, since all four are reasoning systems and the most inert configuration in the entire study is a reasoning one. It is not capability alone, although capability is a precondition: no model below the top commercial tier anywhere in the panel plays the expected-value solution, and the cleanest single contrast we have, between two members of one family and generation that differ only in tier, separates a -1.8-point risk response from a $+120.0$-point one. Capability is therefore necessary and not sufficient, which is the formulation that survives the objection most often raised against nulls of this kind, that the models tested were simply not good enough. Half of the best models available to us were, in this precise sense, not good enough, and it was not because they could not read the game.

The reasoning contrast sharpens the same point into a mechanism. Disabling explicit reasoning in Grok-4.20 raises its contribution from 120.0 to 197.2 at $p = 0.1$ and to 239.4 of 240 in Vietnamese; enabling it returns the model to paying the exact price of the target. Deliberation is worth 105.1 points of 240 in efficiency and nothing at all in risk sensitivity: the

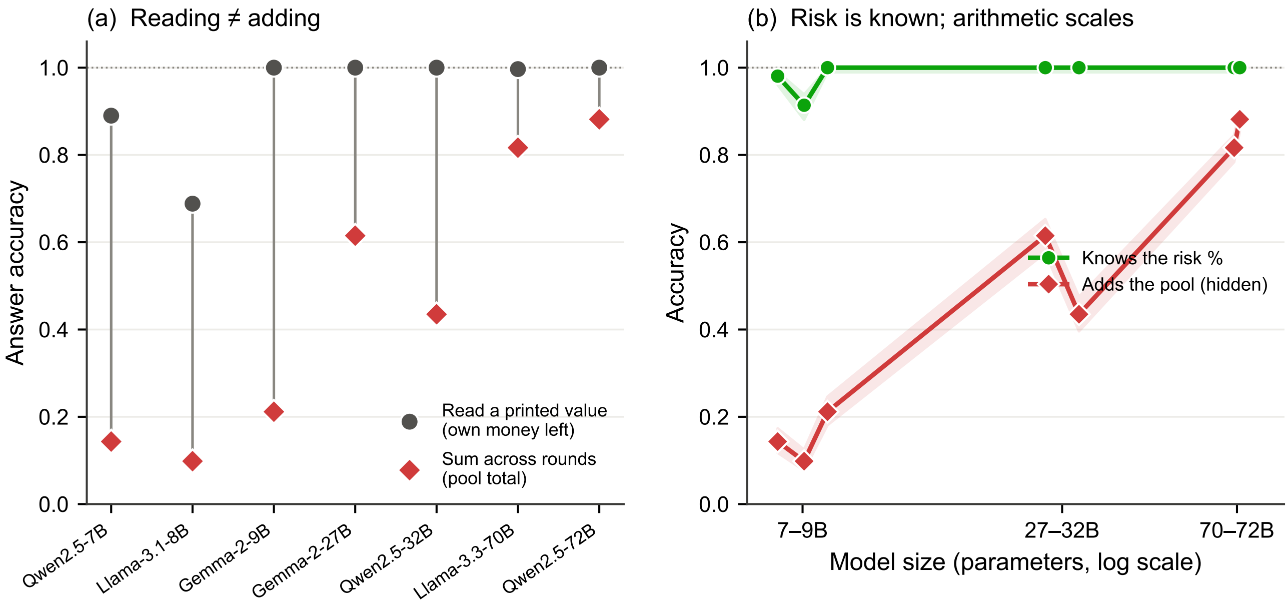

Figure 9. The gap is arithmetic, and it closes with scale. (a) Under the default hidden-pool prompt, accuracy on a value the agent can read straight off the prompt (its own money left, grey circles) versus a value it must sum across rounds (the pool total, red diamonds). Small models can read but cannot add. The same-question control is the A/B in the text: printing the pool total lifts accuracy on that identical question to 92–100%. (b) Knowledge of the catastrophe probability (green) is flat and near-perfect across the size range, whereas the ability to sum the hidden pool (red) rises steeply with scale. Even where both are near-ceiling, behaviour remains risk-insensitive. Bands are Wilson 95% intervals.

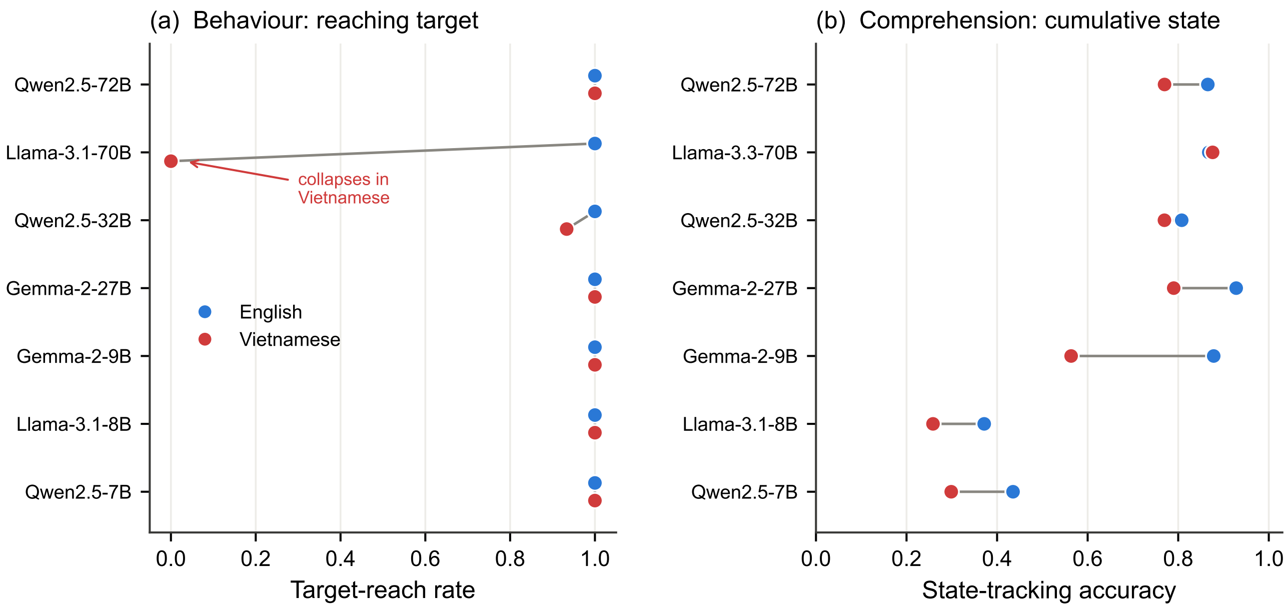

Figure 10. English versus Vietnamese. (a) Target-reach rate in each language for the seven open-weight models, the two points of a pair offset vertically so that coincident values stay visible. Behaviour is identical across languages for five of the seven; Qwen2.5-32B dips slightly (100% to 93.3%) and Llama-3.1-70B collapses from 100% to 0% in Vietnamese. (b) Cumulative-state accuracy in each language. Vietnamese selectively degrades the arithmetic; the largest loss is Gemma-2-9B (31.5 points), while the weakest adders have least left to lose and Llama-3.3-70B is unaffected.

risk slope falls from $+23.6$ to $+0.2$ and the model still contributes 120 at the risk level where the expected-value solution contributes 0. The faculty is spent on the question of how cheaply the goal can be reached, never on the prior question of whether the goal is worth reaching. Read against the open-weight panel, where scaling from 7B to 72B improves cumulative arithmetic from near-chance to near-ceiling [17] while leaving the behavioural prior untouched, the picture is consistent:

Table 1. Headline results per model. Behavioural quantities pooled over risk and language (60 games per configuration; 30 for GPT-5.4-nano): target-reach rate and group contribution (of 240). The risk column gives the change in group contribution from $p = 0.1$ to $p = 0.9$ in English, the study's central quantity. The composition column gives the change from an all-cooperative to an all-selfish group, for the four models run in that study. Comprehension quantities come from the in-situ probes: accuracy at stating the catastrophe probability (*risk knowledge*) and at summing the hidden pool (*cumulative arithmetic*); the commercial panel carries none. Human target-reach rates rise from 0% to 10% to 50% with risk.

		Behaviour		Effect on contribution		Comprehension	
Model	Size or tier	Reach	Contribution	Δ risk (en)	Δ 0→6 selfish	Risk knowledge	Cumulative arithmetic
Open-weight, served locally							
Qwen2.5-7B	7B	100%	232.7	+0.0	−31.8	98.1%	14.3%
Llama-3.1-8B	8B	100%	185.4	+3.4	−94.0	91.4%	9.8%
Gemma-2-9B	9B	100%	128.9	+0.0	−100.4	100%	21.2%
Gemma-2-27B	27B	100%	126.3	+3.8	–	100%	61.5%
Qwen2.5-32B	32B	96.7%	127.6	+1.4	–	100%	43.5%
Llama-3.1-70B	70B	50%	121.0	+0.4	–	100%*	81.7%*
Qwen2.5-72B	72B	100%	132.5	+2.4	–	100%	88.2%
Commercial, hosted							
GPT-5.4-nano	cheap	50%†	136.3†	+7.2	−112.4‡	–	–
Gemini-3.1-Flash-Lite	cheap	100%	120.8	−1.8	–	–	–
Claude-Opus-5	top	100%	124.6	+8.0	–	–	–
Grok-4.20 (reasoning)	top	100%	120.0	+0.2	–	–	–
Grok-4.20 (no reasoning)	top	100%	225.1	+23.6	–	–	–
GPT-5.6-sol	top	65%	87.9	**+118.2**	–	–	–
Gemini-3.1-Pro	top	66.7%	80.1	**+120.0**	–	–	–

*Llama-3.3-70B in the comprehension study. †Pooled over language; the pooled figure is a mixture of 100% reach in English and 0% in Vietnamese and should not be read as a knife-edge strategy (§3(b)). ‡English; the Vietnamese figure is −65.1 from a starting point already below the target. For GPT-5.6-sol and Gemini-3.1-Pro the sub-unity reach rate is the expected-value solution rather than a failure: both reach the target in 100% of games at $p = 0.5$ and $p = 0.9$ and in 0% at $p = 0.1$, where withholding is worth more (§3(c)).

competence and the disposition to act on incentives are separate axes, and moving along the first does not move an agent along the second. Something else does, and our design localises it only to the individual model.

We are equally careful about what the risk-responsive models do not show. They do not reproduce human behaviour; they overshoot it. Human groups reach the target in 0%, 10% and 50% of games as risk rises, whereas Gemini-3.1-Pro and GPT-5.6-sol go 0%, 100%, 100%, a step function at the expected-value threshold. The two failure modes we observe are therefore opposite in direction and equally disqualifying for surrogate use: eleven models cooperate when humans would not, and two coordinate perfectly at a risk level where half of human groups still fail. Nor can we show that these two models compute the expected value. The hosted interface returns the chosen contribution and no deliberation trace, so we can report only that their behaviour coincides, at all three risk levels and in both languages, with what a risk-neutral agent would do. Distinguishing computation from a well-generalised heuristic requires a design that interrogates the agent directly, and it is the natural next study.

The composition study adds a lesson about how such systems fail. The response to composition is close to linear and the threshold is fixed, so whether a group succeeds is determined by where a straight line crosses a horizontal one. This makes collective failure both highly predictable and highly discontinuous: two models with similar per-agent behaviour can have completely different tipping points, and a model can appear perfectly robust across a wide range of compositions and then fail almost completely within one further step. Qwen2.5-7B never failed in 420 composition games not because it resists selfish framing better than the others, since its slope is genuinely negative, but because its intercept is so high that six selfish agents are not enough to bring it down. Robustness, here, is a property of the margin between an intercept and a threshold, not of anything one would call resilience. For anyone deploying a fleet of agents to a shared quantitative goal, the observable per-agent quantities are the intercept and the slope, and these, rather than the observed success rate, determine how much adverse composition the system can absorb.

These findings connect to prior work whose picture is more mixed than a single summary allows. Some studies report LLMs that are markedly more cooperative and more forgiving than humans in repeated dilemmas [8, 9]; others report the opposite temperament, with GPT-4 retaliating unforgivingly after a single defection [10], and most models failing to sustain a shared resource without explicit institutional scaffolding [11]. Our panel suggests that this disagreement may be less a matter of task or of prompt than of which model was to hand: within a single fixed game we obtain both temperaments, and the dividing line runs between models of nominally the same class. The composition result also bears on the practice of persona-conditioning LLMs to stand in for heterogeneous human populations [6, 7]: personas do move behaviour substantially and monotonically, which is encouraging for that practice, but in our data they move it far more reliably than the game's own incentives do for most models, which means a persona-conditioned population may reproduce the distribution of stated dispositions while badly misrepresenting how that population would respond to changing stakes. Methodologically, the study extends the in-situ understanding-check paradigm from the two-player prisoner's dilemma [8] to an N-player threshold game and shows that a single well-chosen probe, can the model state the risk it faces, separates one important confound from a behavioural claim.

Our results carry four practical implications. First, LLM agents are unreliable surrogates for humans in risk-structured collective decisions, and they are unreliable in two distinct ways: most of them predict near-certain cooperation at a level of risk where human groups in fact reach the target only half the time, while the most capable of them predict perfect coordination where humans are divided. Second, an agent's incentive-responsiveness cannot be inferred from its competence, from its vendor, or from its price tier; it has to be measured, and measured on the specific model to be deployed, because two models from the same family and generation differ here by the width of the board. Third, group composition is both the most reliable lever and the most treacherous one, since it moves behaviour linearly and predictably but converts into success or failure through a threshold, so systems can look robust right up until they are not. Fourth, language is not neutral: deploying the same model in a lower-resource language preserved its factual knowledge of the game but degraded its quantitative reasoning, and its behavioural consequences ran in four different directions across four models, from no effect at all to a complete inversion of the outcome. There is no single multilingual correction to apply.

Several limitations bound these conclusions, and the first is severe enough that we place it ahead of the others. Our decision prompt names the equal-split solution. An agent that contributes 2 is therefore consistent with a cooperative disposition, with plain instruction-following, or with anchoring on the only number the prompt offers, and this study cannot tell them apart. The evidence that this is not hypothetical is in our own data: Gemma-2-9B produced a group total of exactly 120.0 in all thirty English games, Gemma-2-27B did the same in Vietnamese, and at the top tier Grok-4.20 with reasoning and both risk-responsive models land on exactly 120 in all but two of the games in which they cooperate at all. Any claim in this paper about cooperative dispositions should be read as a claim about behaviour under a prompt that supplies a focal action; the clean experiment removes the hint and re-runs, and we regard that as the necessary sequel rather than an optional extension. The hint cuts differently for the risk-responsive models, since it makes their behaviour at $p = 0.1$ harder rather than easier to produce: withholding everything requires overriding the focal action the prompt supplies, which is why we regard the 0.0 contributions as the most informative single number in the study.

The remaining limitations are of coverage and implementation. The commercial panel is observed only in the baseline design: it carries no comprehension probes, so we cannot gate its behaviour on understanding as we do for the open-weight models, and no composition study, so the linear-engine description is established at 7–9B and in one cheap commercial model, not at the top tier. Those models are also served through third-party interfaces at provider-default decoding, without per-turn seed control and without any deliberation trace, so their games are unpaired and we observe outcomes rather than process. Their near-determinism cuts both ways: with 16 of the 36 commercial cells showing exactly zero variance across ten games, effects are estimated with unusual precision but the uncensored-cell test that anchors the open-weight analysis cannot be run there at all, which is why the top-tier contrasts are between models rather than within them. Ten repetitions per cell is enough to establish a 120-point difference and not enough to characterise the tail of a rare behaviour. The persona-to-seat permutation is deterministic rather than uniformly random and is not perfectly balanced across seats, so we make no claims about position effects. The groups were homogeneous in model and mute, since all six agents were the same model and could not communicate, whereas human cooperation is shaped by heterogeneity and cheap talk, so our estimates speak to a specific controlled regime rather than to open multi-agent interaction. The catastrophe was a single end-of-game lottery drawn from a schedule shared across conditions, which makes realised disaster rates uninformative and is why we treated reach rate and contribution as the primary dependent variables. Two confounds bound the scaling claim in particular: the 70B and 72B models were served with activation-aware weight quantisation while the smaller ones were not, so scale is not a perfectly clean manipulation, and the Milinski design is old and widely reproduced, so it is likely present in every model's training corpus, meaning an agent may partly be recognising a familiar task rather than reasoning about it afresh. That last concern is sharpest exactly where our new result lives, since the most capable models are the most likely to have absorbed the literature, and it cannot be settled without a novel isomorphic game. Finally, we studied instruction-tuned models only, so the prior we describe is a property of models as deployed and may differ for base models; without base-model controls the attribution to instruction tuning remains a conjecture.

Those limitations define the next steps, and one earlier expectation has already been overturned in a way worth recording. We had expected that forcing explicit accounting, by printing the running total each round, would repair the small models' arithmetic without touching their behaviour, on the reasoning that the risk they must respond to is not the state they must add. That expectation was wrong, and informatively so: printing the total changed contributions more than any incentive manipulation in the open-weight study. The remaining steps are to remove the equal-split hint and re-run; to ask the risk-responsive models directly what they are computing, which is the only way to convert a coincidence with the expected-value solution into evidence of a calculation; to run the composition and comprehension studies at the top tier, where the linear engine and the comprehension gate are both untested; to widen the provider grid, which currently contains no Chinese laboratory because every such model was unavailable through our interface during the run window; and to compare base and instruction-tuned checkpoints, without which the attribution to instruction tuning stays a conjecture.

5. Conclusion

Thirteen instruction-tuned language models, seven open-weight models from 7 to 72 billion parameters across three families and six commercial models spanning two capability tiers at four providers, played a replication of the collective-risk social dilemma. Eleven of them ignored the quantity the game is built around. For those eleven the null is not an artefact of a ceiling, since in the conditions where roughly half of groups succeed and half fail, raising the probability of ruin ninefold moves group contribution by $+0.97$ of a possible 240 ($P = 0.54$), while three manipulations that carry no incentive information at all move the same quantity by one to two orders of magnitude more: printing the running pool total (up to 95.3 points), changing the prosocial composition of the group (31.8–112.4 points), and switching the prompt language (146.8 points, inverting a 100% reach rate to 0%). The other two models overturn any universal reading of that null. Gemini-3.1-Pro and GPT-5.6-sol contribute nothing at all when the catastrophe is unlikely and exactly the target when it is not, a risk effect of $+120.0$ and $+118.2$ points that lifts their realised payoff from 20.0 to 32.0 per agent. Which models do this is predicted neither by provider nor by tier alone: four top-tier models from four laboratories divide two against two, no cheap model ever solves the game, and two members of one family and generation differ by the full width of the board. Enabling explicit reasoning within a single base model buys 105 points of thrift and no risk sensitivity at all. Even the two responsive models are not human, since their reach rate is a step function at the expected-value threshold rather than the graded human curve, so the panel fails as a set of human surrogates in both directions at once. The methodological conclusions are the durable ones. Before a behavioural regularity in LLM agents is attributed to a preference, it should be tested against prompt-side controls that change what the agent is shown without changing what the game makes costly, and any null on an incentive manipulation should be reported alongside a subset of conditions in which that null could have been falsified. And before it is attributed to language models, it should be measured on more than one of them, because in this game the difference between two models of the same generation is larger than the difference between any two conditions we could impose on either.

Ethics.

Data Accessibility.

Authors' Contributions.

Competing Interests.

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